

Task 1: To Parse Append Operator, Open Files in Append Mode, Redirect Streams, Preserve Existing Data, Validate Appends, Test Scenarios

```
GNU nano 7.2 task1.c *
#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>

int main() {
    int fd;

    fd = open("output.txt", O_WRONLY | O_CREAT | O_APPEND, 0644);

    if (fd == -1) {
        perror("output.txt");
        return 1;
    }

    dup2(fd, STDOUT_FILENO);
    close(fd);

    printf("New line added using append mode.\n");

    return 0;
}
```

```
shreyas@Grigori:~$ mkdir -p ~/append_tasks
shreyas@Grigori:~$ cd ~/append_tasks
shreyas@Grigori:~/append_tasks$ nano task1.c
shreyas@Grigori:~/append_tasks$ gcc task1.c -o task1
shreyas@Grigori:~/append_tasks$ echo "Original data" > output.txt
shreyas@Grigori:~/append_tasks$ cat output.txt
Original data
shreyas@Grigori:~/append_tasks$ ./task1
shreyas@Grigori:~/append_tasks$ cat output.txt
Original data
New line added using append mode.
shreyas@Grigori:~/append_tasks$ ./task1
shreyas@Grigori:~/append_tasks$ cat output.txt
Original data
New line added using append mode.
New line added using append mode.
shreyas@Grigori:~/append_tasks$ echo "Shell appended data" >> output.txt
shreyas@Grigori:~/append_tasks$ cat output.txt
Original data
New line added using append mode.
New line added using append mode.
Shell appended data
shreyas@Grigori:~/append_tasks$ ls -l output.txt
-rw-r--r-- 1 shreyas shreyas 102 Sep 27 14:10 output.txt
shreyas@Grigori:~/append_tasks$ wc -l output.txt
4 output.txt
shreyas@Grigori:~/append_tasks$ |
```

Task 2: To Parse stderr Redirection, Redirect Error Streams, Capture Error Output, Verify File Contents, Handle Failures, Test Error Cases

```
GNU nano 7.2 task2.c *
#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>

int main() {
    int fd;

    fd = open("error.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);

    if (fd == -1) {
        perror("error.txt");
        return 1;
    }

    dup2(fd, STDERR_FILENO);
    close(fd);

    fprintf(stderr, "This is an error message.\n");
    fprintf(stderr, "Error stream successfully redirected.\n");

    return 0;
}
```

```
shreyas@Grigori:~/append_tasks$ nano task2.c
shreyas@Grigori:~/append_tasks$ gcc task2.c -o task2
shreyas@Grigori:~/append_tasks$ ./task2
shreyas@Grigori:~/append_tasks$ cat error.txt
This is an error message.
Error stream successfully redirected.
shreyas@Grigori:~/append_tasks$ ls /nonexistent 2> shell_error.txt
shreyas@Grigori:~/append_tasks$ cat shell_error.txt
ls: cannot access '/nonexistent': No such file or directory
shreyas@Grigori:~/append_tasks$ echo "Normal output"
Normal output
shreyas@Grigori:~/append_tasks$ ls /nonexistent 2> error.log
shreyas@Grigori:~/append_tasks$ cat error.log
ls: cannot access '/nonexistent': No such file or directory
shreyas@Grigori:~/append_tasks$ cat error.log
ls: cannot access '/nonexistent': No such file or directory
shreyas@Grigori:~/append_tasks$ chmod 000 error.txt
shreyas@Grigori:~/append_tasks$ ./task2
error.txt: Permission denied
shreyas@Grigori:~/append_tasks$ chmod 644 error.txt
shreyas@Grigori:~/append_tasks$ ls -l
total 56
-rw-r--r-- 1 shreyas shreyas 60 Sep 27 14:12 error.log
-rw-r--r-- 1 shreyas shreyas 64 Sep 27 14:11 error.txt
-rw-r--r-- 1 shreyas shreyas 102 Sep 27 14:10 output.txt
-rw-r--r-- 1 shreyas shreyas 60 Sep 27 14:12 shell_error.txt
-rwxr-xr-x 1 shreyas shreyas 16128 Sep 27 14:08 task1
-rw-r--r-- 1 shreyas shreyas 360 Sep 27 14:08 task1.c
-rwxr-xr-x 1 shreyas shreyas 16176 Sep 27 14:11 task2
```

```
-rwxr-xr-x 1 shreyas shreyas 16128 Sep 27 14:08 task1
-rw-r--r-- 1 shreyas shreyas 360 Sep 27 14:08 task1.c
-rwxr-xr-x 1 shreyas shreyas 16176 Sep 27 14:11 task2
-rw-r--r-- 1 shreyas shreyas 422 Sep 27 14:11 task2.c
shreyas@Grigori:~/append_tasks$ cat error.txt
This is an error message.
Error stream successfully redirected.
shreyas@Grigori:~/append_tasks$ cat error.log
ls: cannot access '/nonexistent': No such file or directory
shreyas@Grigori:~/append_tasks$ rm -f task1 task2 output.txt error.txt shell_error.txt error.log
shreyas@Grigori:~/append_tasks$ ls
task1.c task2.c
shreyas@Grigori:~/append_tasks$ |
```